Question	Answers	Extra information	Mark	AO / Spec Ref.
01.5		allow conversion to mg at any point in calculation		AO2 4.3.3.1
	(0.6 × 1.2 =) 0.72		1	
	(0.72 × 2 =) 1.44		1	
	1.44 × 1000	allow conversion of a correct mass calculated for berries / roots	1	
	1440 (mg)	allow a correct answer using an incorrect conversion factor of a multiple of 10	1	
	alternative route (2 × 1.2 =) 2.4 (1) (2.4 × 0.6 =) 1.44 (1) 1.44 × 1000 (1) 1440 (mg) (1)			

Question	Answers	Extra information	Mark	AO / Spec Ref.
01.6	any **one** from:		1	AO1 4.3.3.1 4.3.1.2 4.3.1.4
	• lack of chlorophyll	allow lack of chloroplasts		
	• lack of magnesium (ions)	allow lack of mineral (ions) **or** named example such as iron / nitrate / potassium / zinc (ions) ignore lack of nutrients / nitrogen allow lack of water allow incorrect pH of soil allow acidic / alkaline soil allow (named) infection allow lack of light allow other correct suggestions		

Question	Answers	Extra information	Mark	AO / Spec Ref.
01.7	any **two** from: • to check toxicity • to check dosage • to check its efficacy	 allow to check it is safe allow to check for side effects allow to check it is not poisonous / dangerous / harmful allow to check how much is needed allow to check it works allow to check it does not interact with other drugs	2	AO1 4.3.1.9

Question	Answers	Extra information	Mark	AO / Spec Ref.
01.8	(writers / companies may get) financial gain **or** (competitor may suffer) financial loss	ignore because the report has not been peer reviewed ignore have not used a double-blind trial	1	AO3 4.3.1.9

Question	Answers	Extra information	Mark	AO / Spec Ref.
01.9	have the claims peer reviewed		1	AO1 4.3.1.9

Total Question 1	**13**

Question 2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.1	A		1	AO1 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.2	C		1	AO1 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.3	right atrium		1	AO1 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.4	any **two** from: • (artery) has a thick**er** muscle (tissue) • (artery) has a thick**er** elastic (tissue) • (artery) has a narrow**er** lumen • (artery) does not contain valves	allow converse if clearly referring to a vein if neither mark awarded allow **1** mark for artery has a thick**er** wall allow description of lumen	2	AO1 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.5	as the percentage of the (coronary) artery that is blocked increases, blood flow decreases	allow converse allow the greater the blockage, the less blood flows allow negative correlation **or** inversely proportional allow as one increases, the other decreases	1	AO2 4.2.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.6	scale on y-axis	must take up at least 50% of axis	1	AO2 4.2.2.2
	all points plotted	allow 3 or 4 correct plots for **1** mark allow a tolerance of ± ½ small square	2	
	correct curved line of best fit	ignore line joined point to point with straight lines ignore extrapolation	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
02.7	correct answer from student's line in **Figure 3**	allow a tolerance of ± ½ small square if no line drawn on **Figure 3**, allow a value from 18 to 24 (cm^3/minute)	1	AO3 4.2.2.2

Question	Answers	Mark	AO / Spec Ref.
02.8	**Level 3**: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5-6	AO1 AO2 4.2.2.2 4.2.2.4 4.4.2.1 4.4.2.2
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3-4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1-2	
	No relevant content.	0	
	Indicative content: • reduced blood flow to heart (muscle / tissue / cells) • (so) less oxygen to heart (muscle / tissue / cells) • (so) less glucose to heart (muscle / tissue / cells) • (so) less (aerobic) respiration (in heart / body cells) • (more) anaerobic respiration • (so) less energy (released) • (so) less muscle contraction • (so) less blood / oxygen / glucose around the body (from heart) **or** slower flow of blood / oxygen / glucose to body (from heart) • less carbon dioxide removed from body (muscle / tissue / cells) • (resulting in) breathlessness • (resulting in) tiredness • (anaerobic respiration causes) production of lactic acid • (build-up of lactic acid) causes muscle fatigue / pain **or** chest pain For **Level 3**, students must explain the effect of reduced oxygen / glucose on respiration **or** energy release and its consequence		